

Comparative Technical Analysis of Image Processing and Computer Vision

**ITA0506 – COMPUTER VISION
TECHNICAL ASSESMENT CO1-AT3**

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Introduction to Image Processing and Computer Vision

- Image Processing improves the quality of digital images.
- Computer Vision enables computers to understand image content.
- Both technologies use digital images as input.
- Image Processing is usually the first step before Computer Vision.
- They are widely used in AI, healthcare, robotics, and autonomous vehicles.

Process: 📷 Camera → 🖼️ Image Processing → 🤖 Computer Vision → 🚗 Decision.



Image Processing

What is Image Processing?

- Enhances image quality using digital algorithms.
- Removes noise and improves brightness and contrast.
- Detects edges and sharpens image details.
- Used in photo editing, medical imaging, and satellite images.
- Produces an improved image but does **not** understand its content.



Computer Vision

What is Computer Vision?

- Extracts useful information from digital images.
- Detects, recognizes, and classifies objects automatically.
- Uses Artificial Intelligence and Machine Learning techniques.
- Applied in face recognition, self-driving cars, and surveillance.
- Produces decisions or predictions instead of only improving images.



Comparison of Image Processing and Computer Vision

Comparison

- Goal: Image Processing improves images, while Computer Vision understands them.
- Output: Enhanced image vs. meaningful information or decision.
- Technology: Traditional algorithms vs. AI and Deep Learning.
- Applications: Photo editing vs. object recognition and automation.
- Image Processing supports Computer Vision by improving image quality.



Applications and Conclusion

- Healthcare: Image enhancement and disease detection.
- Autonomous Vehicles: Road image processing and obstacle detection.
- Security: CCTV enhancement and face recognition.
- Agriculture: Crop image enhancement and disease identification.
- **Conclusion:** Image Processing improves images, while Computer Vision converts images into intelligent decisions.